Auto Ticket Classification using Flow Designer

Technology Track

1. PURPOSE

This document lists the technologies chosen to build the Auto Ticket Classification solution and explains the role of each one. The choice follows the functional and non-functional requirements, and the goal is to automate triage using tools that are already part of the service desk platform.

2. TECHNOLOGY TRACK OVERVIEW

Layer	Technology / Component	Role in the project
Platform	Enterprise ITSM platform (cloud, low-code)	Hosts the service desk, ticket tables, users, groups and the automation engine.
Automation	Flow Designer	Runs the classification and routing logic with no manual triage.
Rule logic	Keyword and decision rule table, flow conditions	Stores keyword to category, priority and group mapping that admins can edit.
Data	Incident, Request Item, Assignment Group and SLA Definition tables	Hold the ticket data and reference data read and updated by the flow.
Notification	Platform notification engine (email and in-app)	Alerts the assigned resolver group as soon as the ticket is routed.
SLA	Platform SLA engine	Attaches response and resolution timers based on priority.
Reporting	Dashboards and reports	Show routing accuracy, re-assignment rate and SLA compliance to the IT Manager.
Testing	Development instance, flow test run, sample incident dataset	Verify every rule and route before it goes live.

3. TECHNOLOGY ARCHITECTURE

The technology track has four layers. Tickets enter through the input layer, Flow Designer processes them, the data layer supplies rules and reference data, and the output layer delivers the routed ticket, alerts and reports.

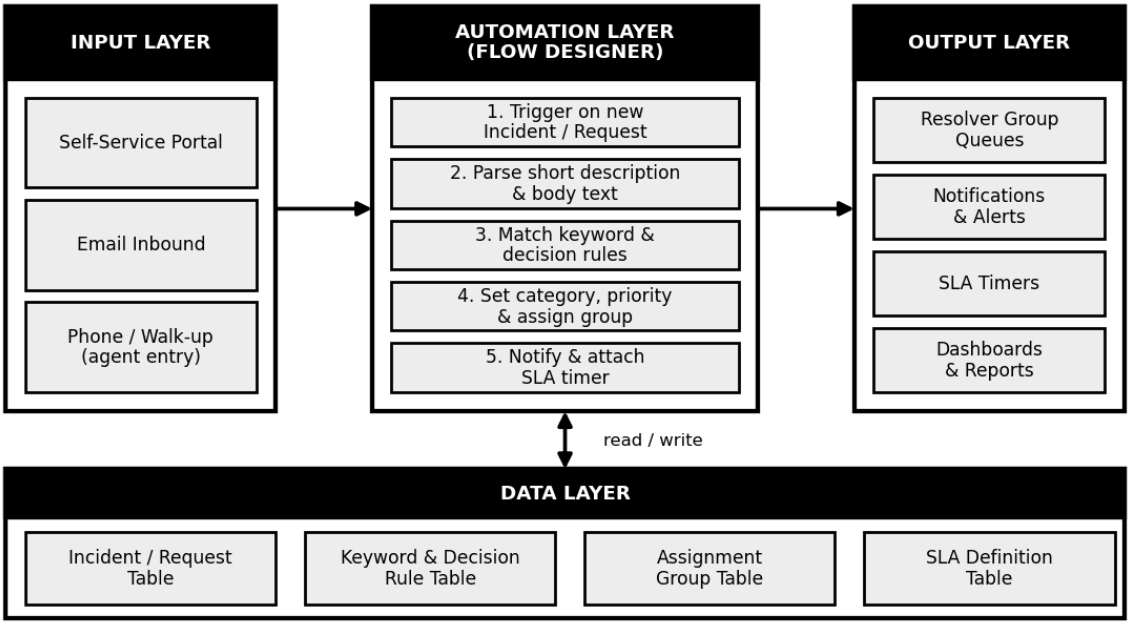


Figure 1: Layered technology architecture

4. FLOW DESIGNER COMPONENTS USED

Flow Designer Component	Use in this project
Trigger	Record created on the Incident and Request Item tables starts the flow for every new ticket.
Conditions and logic	If / else branches check keyword matches and decide the category, priority and group.
Look up record	Reads the matching row from the keyword and decision rule table.
Update record	Writes Category, Subcategory, Urgency, Priority and Assignment Group to the ticket.
Notification action	Sends an alert to the assigned resolver group.
Subflow	A reusable classification subflow that can serve both Incident and Request flows.
Error handling	Sends any failed or unmatched ticket to the default review queue for a Tier-1 Agent.
Execution history	Shows every flow run, so a wrong routing can be traced and fixed.

5. WHY THIS TECHNOLOGY

Requirement	How the technology meets it
Instant classification (under 10 seconds)	Event-driven trigger runs the moment the ticket is created.
Easy rule changes without code	Rules live in a table and flows are built with a visual, low-code designer.
Consistent decisions	The same rule table drives every ticket, so identical issues get identical results.
No lost tickets	Error handling and the review queue catch anything the rules cannot match.
Fast delivery	Uses features already inside the platform, so no new tools or integrations are needed.

6. ENVIRONMENTS AND TOOLS

Environment / Tool	Purpose
Development instance	Build and change the flow, rule table and notifications safely.
Test instance	Run the standard incident dataset and check accuracy and speed.
Production instance	Live service desk where the tested flow is activated.
Change packaging	Move the flow and rule table from development to test and production as one controlled package.
Sample dataset	Incidents covering VPN, password, hardware, database and security issues, plus vague ones such as "System error".

7. SKILLS REQUIRED

- **Flow Designer:** building triggers, conditions, actions and subflows.
- **ITSM knowledge:** understanding incidents, requests, categories, priorities and SLAs.
- **Rule design:** writing clear keyword and decision rules that do not overlap.
- **Testing:** preparing sample data and checking accuracy, speed and fallback behaviour.

8. TECHNOLOGY RISKS AND MITIGATION

Risk	Impact	Mitigation
Vague ticket text	No keyword match	Review queue for Tier-1 Agents; guided fields on the portal.
Rules become outdated	Accuracy drops over time	Monthly rule review using the manager report.
Overlapping keywords	Wrong group assigned	Rule order and priority; test each rule with sample data.
Flow failure at peak load	Tickets left unassigned	Error handling, monitoring and execution history checks.

9. CONCLUSION

A low-code, event-driven technology track built on Flow Designer and the existing ITSM platform is enough to meet every requirement. It removes manual triage, keeps rules easy to maintain, and can be delivered quickly without extra tools or integrations.